

EEE - 4483

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Section : 1A

Expt. Name : Study of RTL NOR Gate and TTL NAND Gate

Expt. No : 02

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Objective:

The goals of the experiment are

1. To study and understand the working of an RTL NOR Gate.
2. To analyse a TTL NAND Gate circuit with a totem-pole output stage.

List of Equipments:

1. DC power Supply
2. Breadboard
3. Resistors (5 pieces)
4. Diode (1 piece)
5. Transistors (BJT) - 5 pieces
6. Connecting Wires
7. Multimeter

Data Tables:

RTL NOR Gate:

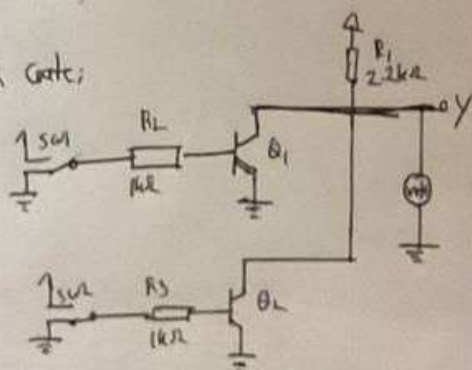
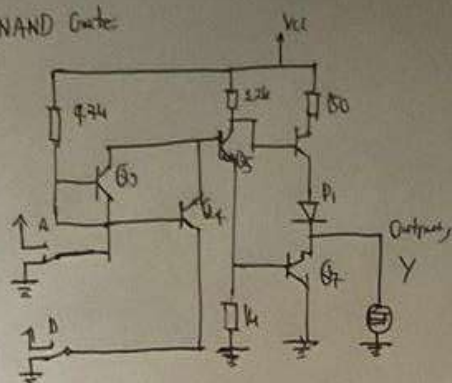


Fig. 2(a). NOR Gate

Input A	Input B	Output Voltage (V)	Logic Output Level
0	0	5.05V	1
0	1	11.9mV	0
1	0	10mV	0
1	1	9mV	0

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NAND Gate



Input A	Input B	Output Voltage	Logic Output (or)
0	0	4.6V	1
0	1	4.4V	1
1	0	4.53V	1
1	1	10mV	0

Justifications:

i) RTL NOR Gate: The gate uses resistors at the input and a transistor at the output. to perform the logic operation, if either input is HIGH, the base of the transistor is forward biased, turning it on and pulling the output low. The output is high only when both inputs are low. This behavior matches NOR Gate logic.

Result: Output was HIGH only when both inputs are LOW.

ii) TTL NAND Gate: This circuit uses multiple transistors where the first stage detects both inputs and the totem-pole output configuration, i.e., one transistor pulling up and another pulling down which provides strong output signals. The NAND gate outputs LOW only when both inputs are HIGH. In other ^{cases} ~~cases~~, the output stays HIGH.

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which is confirmed by the data.

Answers to Report Questions:

1. The measured output for all report combinations are match the expected truth tables of NOR and NAND Gates.

The RTL NOR Gate only gives a high output when both inputs are low.
The TTL NAND gate gives a low output only when both inputs are high.
This confirms correct operation based on theoretical logic gate behavior.
2. The features and advantage of TTL gates are:
 - i) High Speed: TTL gates switch faster than earlier logic families like RTL or DTL.
 - ii) Totem pole output: Allows faster charging / discharging of the output load.
 - iii) Low Power Consumption: Consumes less power than earlier families.
 - iv) Good Noise Margin: More Reliable signal handling.
 - v) Wide Availability: Easily available with standardised ICs.

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3. The diode D_1 is used to prevent backflow of current from the output stage. It helps to ~~pr~~ protect the transistor and control ~~current~~ current flow in the circuit, especially during switching transitions, improving the speed and performance of the output stage.

4. ~~The~~ the totem pole configuration consists of two transistors stacked vertically:

- i) The upper transistor pulls the output ~~High~~ High when turned on.
- ii) The lower transistor pulls the output Low when turned on.

This arrangement allows faster switching and better current driving capability, as it actively drives the output both High and Low, instead of relying on pull-up resistors.

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Discussion:

In the experiment, we built and tested two types of logic gates using BJTs: an RTL NOR Gate and a TTL NAND Gate. We saw how the RTL uses resistors and transistors to perform the NOR logic. It only gives a High output when both inputs are low. For the TTL NAND Gate, we learned about the totem pole design that helps the circuit respond faster. It gives a Low output only when both inputs are high. The diode in this gate helps control the current and protect the components.